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e-Payments:

“BLOCKCHAIN Essentially Creates A Trusted Environment To Operate On Multi-Company Supply Chain Ecosystems”

The Internet of Things, cognitive manufacturing and connected systems have one thing in common—all work on data that has been collected. Cognitive tools start to analyse all things that have been brought in. So how does it work? Bruce Anderson, electronics industry global managing director at IBM, speaks with Shanosh Kumar from EFY

Q. Please give an example of a cognitive machine learning to use natural language?

A. When IBM Watson played Jeopardy, early versions were only getting half of the questions right. It later managed to dramatically improve with almost no training on data. How? The learning aspect is critical. Human beings, as experts, came in to help this cognitive system understand the taxonomy of the problem being worked on.

Q. Can you explain the role of taxonomy here?

A. The literal and logical meaning of words is understood by the cognitive system after it builds a lot of context to understand domain knowledge. This is very natural for a human being, but not so much for a computer. Engineers need to realise that, with cognitive, half the problem is about capturing the right data, while the other half is about training the system using acquired domain knowledge. They must use that corpus of data to interrogate the system using natural language.

Q. Coming back to the electronics industry, where do blockchains fall in the world of connected devices?

A. The electronics industry is very complex and multi-layered, and companies that interact with each other are sending transactions back

and forth. If minor problems occur, companies often need to spend millions of dollars to resolve supply chain issues. Blockchain essentially creates a trusted environment to operate on multi-company supply chain ecosystems, as it helps eliminate friction that can occur due to lack of trust, especially when you consider the huge amount of verification and agents involved with transactions.

Q. Are blockchains all about evolving into an automated, secure methodology of handling business transactions?

A. In certain cases, automation of tasks can be used to eliminate manual pieces. In the most extreme cases, it can eliminate human tasks itself. If you think about automation using blockchain, tasks could be done intelligently. Blockchain may act as a ledger between the two companies, where data from a variety of sources need to be retrieved even for something as simple as payment processing.

Now, all these tasks could be supported in a blockchain application. In some cases, human beings are still going to be involved for setting up the blockchain system and to make appropriate approvals and verifications.

Q. How does cognitive technology emerge with blockchain, and how deep does it go working hand in hand?

A. Putting cognitive analytics on top of blockchain provides the capability to look for emerging patterns and trends. There are opportunities to detect fraud, discrepancies and root cause analysis to create a trusted source of data that form the core of business processes. Cognitive analytics and blockchain are not really tightly mixed. Cognitive analytics is about studying patterns from data that has been vetted by blockchain.

More importantly, data in blockchain ledger can be structured or unstructured, requiring cognitive technology for analysis. The source of information could come from a manufacturing sector, traditional business system or a variety of other areas.

Q. How do these two technologies look at data?

A. Blockchain looks at a massive amount of fairly-organised information. The cognitive layer tries to understand the data layer formed by the blockchain and builds the indices that are necessary to answer questions in natural language. This is the way cognitive systems understand, reason and learn.

With data in the blockchain from many other data sources, applications can be built on top to literally interrogate the system using natural language, just like how humans talk to each other. **EFY**